TECHNICAL DESIGN & SOLUTION DOCUMENT

Sub-Millisecond IBM Z Real-Time AI Triage & Emergency Response System

Track: Real-Time AI for Critical Decisions — **Target Architecture:** IBM z16 (Telum AI Accelerator)

1. Executive Summary & Problem Definition

Problem Statement

Emergency medical and response teams face critical delays due to high latency and security risks in cloud AI. Sub-millisecond IBM Z inferencing is needed for real-time triage and dynamic response.

Core Challenges

- **Network Latency & Jitter:** Public cloud AI architectures introduce $100ms$–$500ms+$ network delays, making real-time telemetry processing unsafe for split-second medical triage.
- **Data Privacy & Compliance:** Transmitting streaming patient vitals across public networks creates security risks and violates strict healthcare compliance regulations (HIPAA, GDPR).
- **Single Point of Failure:** Emergency dispatch and triage systems demand 99.999% availability. Cloud connection failures during major disaster events stall emergency response pipelines.

2. Proposed Solution Architecture

We propose an enterprise-grade, end-to-end real-time decision-support system running natively on **IBM Z infrastructure**. By co-locating transactional workflows and AI inferencing on the same system, the platform processes streaming vitals and incident telemetry with **sub-millisecond latency**.

System Architecture Flow

[**Streaming Telemetry**] (Ambulance GPS, MIMIC-IV Vitals, 911 Logs)
$\longrightarrow$ [**IBM z/OS Data Streaming**] (Apache Kafka / Event Streams)
$\longrightarrow$ [**IBM Telum Integrated AI Accelerator**] (Sub-Millisecond On-Chip Inferencing)
[**Real-Time Patient Triage**] (NEWS2 Risk Score Analytics & Instant ICU Alerts)
[**Dynamic Resource Allocation**] (Optimal Route Optimization & Bed Reservation)

Key Pillars

- **On-Chip Acceleration:** Executes deep learning models on the **IBM Telum processor's AI accelerator** directly alongside core data streams with zero PCIe bus latency.
- **Zero Trust Security:** Uses IBM Z Pervasive Encryption and CPACF hardware acceleration to protect data in flight, at rest, and in memory during model execution.
- **High Reliability:** Operates on LinuxONE / z/OS clusters to maintain zero downtime during emergency load spikes.

3. Data Pipeline & Specifications

Dataset	Source Type	Key Features / Metrics	Pipeline Role
Real-Time Healthcare	MIMIC-IV Streams	Heart rate, SpO_2, blood pressure, respiratory rate	Dynamic NEWS2 risk scoring & deterioration detection.
Emergency Logs	911/108 Logs	Dispatch timestamps, severity codes, location data	Queue priority classification & dispatch optimization.
Geospatial Telemetry	Live GPS Streams	Traffic speeds, ambulance coordinates, ETA metrics	Dynamic route optimization & congestion avoidance.

Execution Pipeline Steps:

1. **Ingestion:** Apache Kafka on z/OS ingests continuous telemetry streams from field units.
2. **Transformation:** In-memory streaming computes rolling 30-second moving averages for feature extraction.
3. **Inference:** Pre-trained ONNX neural network models execute on the IBM Telum AI accelerator ($< 1ms$).
4. **Action:** High-risk alerts are delivered to ICU dashboards and paramedic mobile endpoints.

4. Target Audience & Impact Analysis

1. **Emergency Medical Services (EMS) & First Responders:** Provides field paramedics with real-time risk scores and route optimization to shorten critical transit windows.

2. **Hospital Intensive Care Units (ICUs) & Triage Teams:** Delivers early-warning patient deterioration alerts to enable proactive ICU bed and specialist preparation before arrival.
3. **Disaster Management & Public Safety Agencies:** Equips regional command centers with real-time situational awareness and predictive dispatch during large-scale emergencies.

5. Practical Hackathon Implementation Roadmap

This structured 4-phase execution strategy ensures a functional end-to-end prototype delivered within datathon timelines:

Phase	Core Deliverables	Technical Stack / Tools
Phase 1: Data Prep	Process MIMIC-IV sample vitals and synthetic GPS streams into JSON/CSV event queues.	Python, Pandas, Synthetic Data Generator
Phase 2: Model Training	Train lightweight XGBoost & ONNX classifiers for NEWS2 risk scoring. Quantize model for fast inferencing.	Scikit-Learn, ONNX Runtime, PyTorch
Phase 3: IBM Z Simulation	Simulate low-latency batch inferencing environment using z/OS containerized runtime / LinuxONE trial.	Docker, LinuxONE Community Cloud, ONNX
Phase 4: Dashboard	Build a real-time dashboard displaying patient risk alerts, ETA map routes, and system latency metrics.	Streamlit / Gradio, Folium Maps

Winning Strategy Focus

- **Proof of Latency Advantage:** Benchmark ONNX model execution times to prove sub-millisecond inferencing capabilities against standard cloud APIs.
- **Clear Live Demonstration:** Present a working Streamlit dashboard showing simulated ambulance streams triggering high-priority ICU alerts in real time.
- **Enterprise Realism:** Highlight IBM Z architectural alignment (z16 Telum accelerator, z/OS Kafka, hardware encryption) to show production feasibility.

6. Security, Privacy & Compliance

- **Privacy-Preserving AI:** Keeps raw patient biometric data secured in localized enclaves while syncing model updates safely.
- **Hardware Cryptography:** Leverages IBM Z CPACF hardware acceleration to encrypt streaming data at wire speed with zero performance overhead.
- **Audit Trail:** Maintains logging on z/OS for full regulatory compliance (HIPAA/GDPR) and clinical decision auditing.